AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 36 · STUDENT LESSON

# Expo and Defense

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

## Show the trace. Compare honestly. Defend the limits.

Evaluate the fixed activity adviser, preserve actual evidence and explain what its results support before planning the next step.

---

**Keep one habit:** Archive the candidate → record both methods → defend a bounded conclusion.

---

MAIN LESSON · SECTION 1

## What can your project evidence actually defend?

> Explain results, failures and limits rather than relying on presentation polish.

### Start with an evidence-based defence

Your strongest defence is an inspectable trace, an honest comparison and a conclusion limited to the evidence. Keep the evaluated instructions unchanged even when a result surprises you.

## Recall when a seen case stops being unseen

You change a candidate after seeing a final test fail. Can the same case be an unseen final test of the new version? Explain.

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

## Your learning goals

- I can evaluate a saved candidate and a manual **baseline** fairly.
- I can defend a conclusion using results, failures and limits.
- I can explain the rules, privacy choices and **human control** in my project.

▼ **Earlier wording, alternatives and picture transcripts**

### Your learning target

### Remember before reading

Without opening the old lesson, explain one key idea from Improve One Cause. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Test Normal Cases. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Choose **Input**s and **Output**s. Give an example and a mistake that the idea helps you avoid.

Understanding is shown by explanation, **transfer**, testing, and limits.



LEARNING TARGET Defend a sorter with 10/12 routine successes and 2/4 mixed-material successes. Report overall and separate results, show one failure and propose a justified next step. What claim would overstate the **evidence**?

Learner name:

### Readiness for the connected explanation

### This week’s end goal

By the end, I can defend a project using measured results, limits and an honest **baseline** comparison.

MAIN LESSON · SECTION 2

# Save the candidate before final inputs are revealed

> Evaluate the activity adviser under its existing contract.

## Fix the saved version and retain known problems

This week you will evaluate your activity adviser and explain what its **evidence** supports. A project with an honestly reported failure can show strong understanding.

Save your candidate instructions, version label, **input** rules, development records and known problems before your teacher reveals the final requests and catalogue. Keep the instructions unchanged during this evaluation.

### Read the unchanged input and matching rules

The agreed rule remains: validate whole available minutes from 0 to 30 and Quiet required Yes or No. No allows quiet or nonquiet activities.

Start at the first row and suggest the first activity whose time is no more than available time AND whose quiet label satisfies the request. Explain the choice. If all rows fail, report no match.

Invalid or missing **input** needs clarification before searching. Clear the previous **output** for each run.

## Use the assigned catalogue as input, not a reason to rewrite instructions

Your teacher will supply a fresh, labelled catalogue in the same format. Its values are test **input**s, not permission to rewrite your instructions. All materials are assumed available.

The manual **baseline** checks every row against the same requirements, then selects the first fit. Both methods advise only; neither books, publishes or chooses on a person’s behalf.

## ▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

### Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY A team presents the Classroom Sorter, traces a new item live, shows test **evidence**, explains one failure, and states when a human must decide.

### What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

### A closer explanation

# Record both methods before checking the key

> Use the same requests, catalogue and full-pass criteria.

## Follow the final-evaluation record

### From saved version to supported claim

| BEFORE THE RUN | DURING THE RUN | AFTER THE RUN |
| --- | --- | --- |
| Save instructions and label | Use the assigned data | Keep results and traces |
| Keep final cases reserved | Run both methods fairly | Compare full passes |
| State rules and limits | Record actual outputs | Explain failures and limits |

Changing instructions during evaluation creates a different version. Keep evidence for the original version.

## Keep observations separate from the teacher's expectations

Use the same final requests, catalogue, requirements and full-pass rule for both methods. Record the candidate **output** and reason before looking at the expected answer.

Record the **baseline** separately; do not copy the candidate result into it. Your teacher holds the answer key.

A full pass needs both the required **output** and a valid reason, including why earlier rows fail when a later row is selected.

## Record versions, reasons, failures and unrun cases

- For each request record its ID, **input**s, catalogue label, candidate label, candidate **output** and reason, **baseline output** and reason, and both results after checking. Keep a trace for any failure. Mark unrun cases explicitly.

  Do not silently correct an **observation**. If a person misreads an instruction, record that execution error separately from a defect in the written instruction.

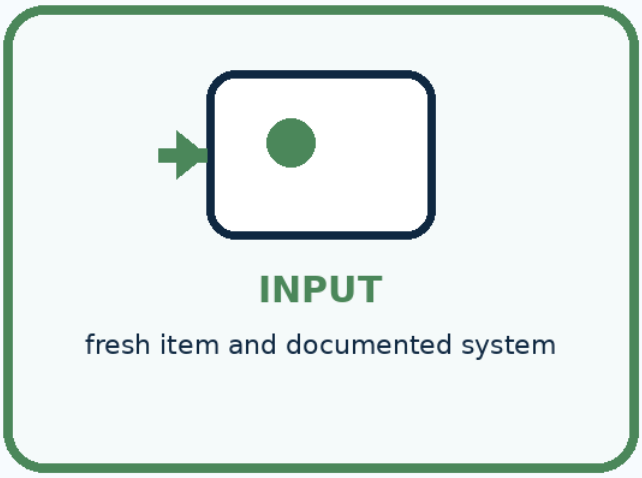
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

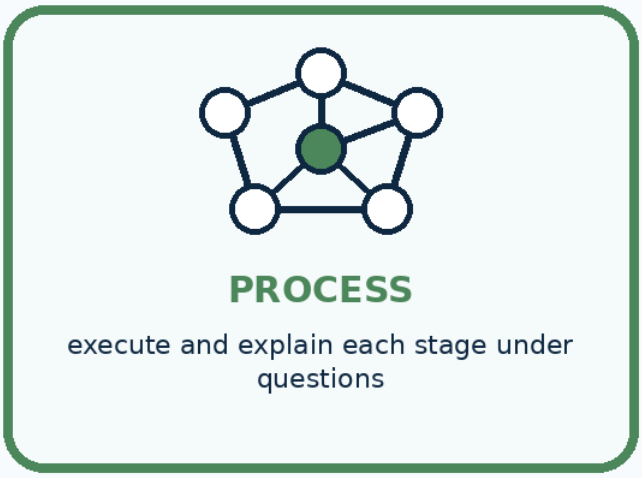
### ▼ Earlier wording, alternatives and picture transcripts

VISUAL **MECHANISM**

**Input → Process → Output**

Follow the information instead of imagining magic inside the machine.









**INPUT** fresh item and documented system

**PROCESS** execute and explain each stage under questions

**OUTPUT** result, **evidence**, **limitation**, and responsible next action

CORE IDEA Understanding is shown by explanation, **transfer**, testing, and limits.

UNDER THE HOOD A **defense** evaluates **transfer**able understanding, not presentation polish.

Strong **evidence** includes system architecture, version history, planned tests, failures, controlled improvement, fairness/privacy boundaries, and independent response to a novel case.

Draw a fresh example with all three stages:

**A closer explanation**

**Connect the picture and the explanation**

---

MAIN LESSON · SECTION 4

## Compare presentation aids with evidence of understanding

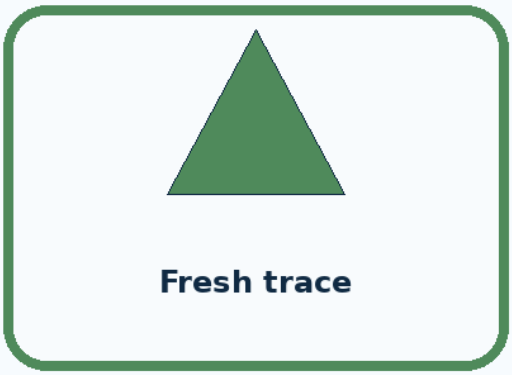
> A fresh trace, failure record and human boundary support a claim that a poster alone cannot.

## Pictures paired with their evidence




**The case to investigate**

CASE 1 — Polished poster Clue: communication aid. Meaning: not proof alone.




**The case to investigate**

CASE 2 — Fresh trace Clue: unrehearsed application. Meaning: **transfer evidence**.




**The case to investigate**

CASE 3 — Failure log Clue: honest testing record. Meaning: engineering **evidence**.




**The case to investigate**

CASE 4 — Human gate Clue: responsible boundary. Meaning: use condition.

## Discuss the confusing pair

- Which two pictures are easiest to confuse? What **evidence** separates them?

### Keep the limit in view

BOUNDARY Confidence, decoration, or memorized vocabulary cannot replace a fresh trace and **evidence**-based answers.

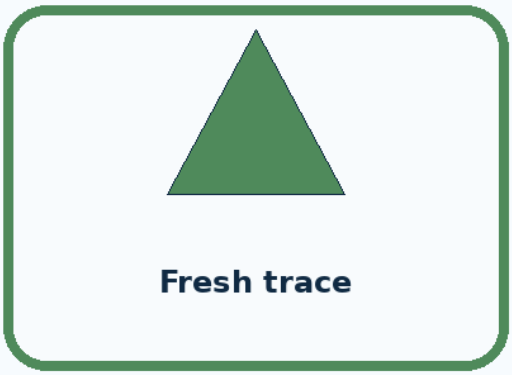
▼ **Earlier wording, alternatives and picture transcripts**

PICTURE GALLERY

**Four Cases, Four Clues**

Use the visual cue, then read the **evidence** carefully.









CASE 1 — Polished poster Clue: communication aid. Meaning: not proof alone.

CASE 2 — Fresh trace Clue: unrehearsed application. Meaning: **transfer evidence**.

CASE 3 — Failure log Clue: honest testing record. Meaning: engineering **evidence**.

CASE 4 — Human gate Clue: responsible boundary. Meaning: use condition.

### Change one thing in the pictured case

Change: Someone adds the two unrun cases to the denominator as failures.

**Prediction**: That reports a different measure, 8/12 completed successes out of planned cases.

Reason: It cannot be called the original attempted-case pass rate; distinguish completion from evaluated correctness.

MAIN LESSON · SECTION 6

# Audit the separate exhibition-board comparison

> This fictional ten-attempt record differs from the earlier eight-attempt illustration and your actual evaluation.

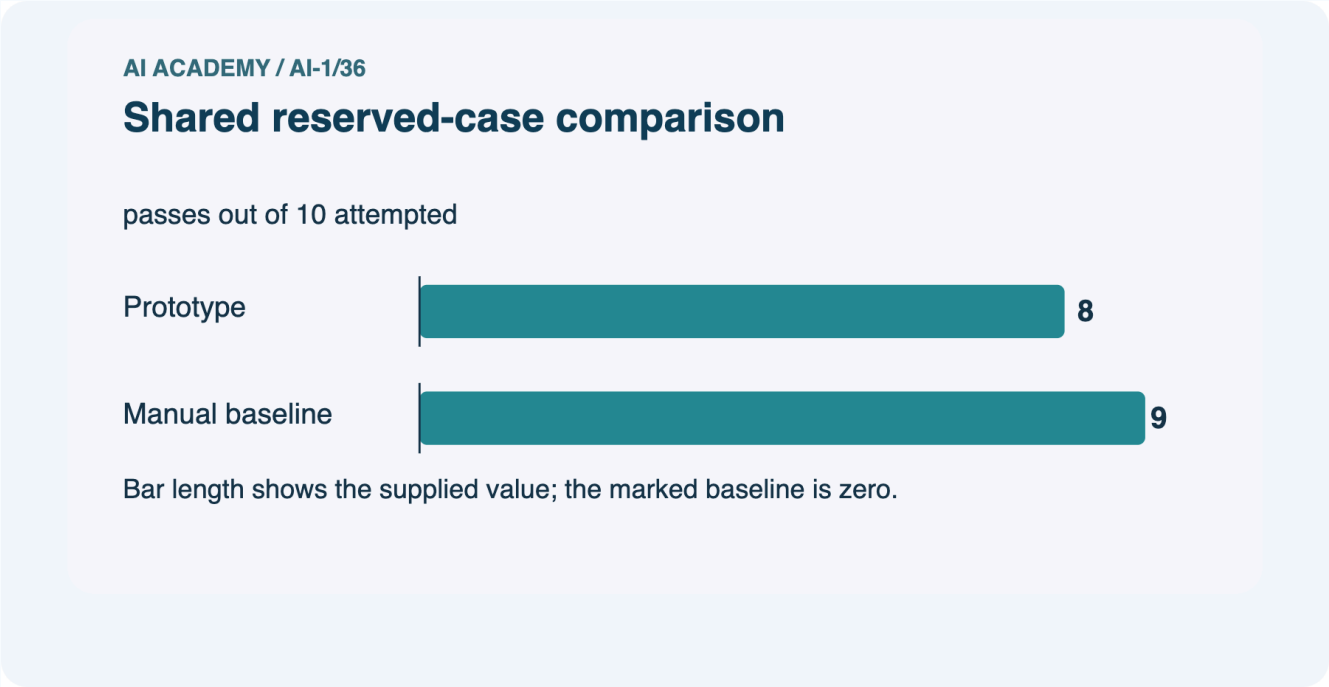
## Read the ten shared attempts and two unrun plans

A fictional paper sorter team prepares an exhibition **evidence** board.

Both prototype and manual **baseline** attempt the same 10 reserved cases. Prototype passes 8; **baseline** passes 9.

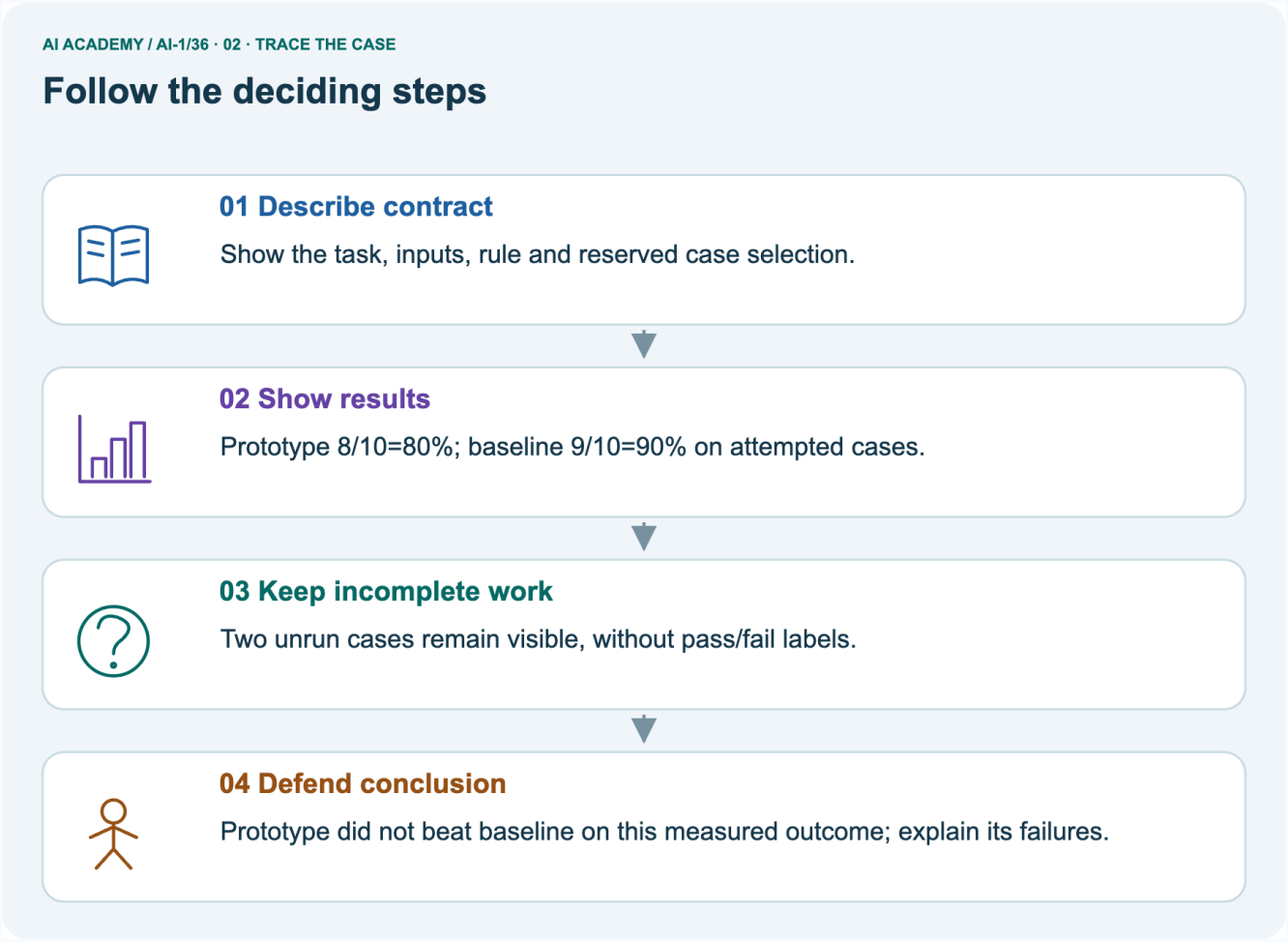
Two additional planned cases are unrun. No timing or pupil-satisfaction measures were collected.

## Inspect the labelled shared-case pass counts




Follow the labelled picture with the matching explanation. Open the vector diagram

## Trace the result and limited conclusion




Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Describe contract | Show the task, inputs, rule and reserved case selection. |
| 2. Show results | Prototype 8/10=80%; baseline 9/10=90% on attempted cases. |
| 3. Keep incomplete work | Two unrun cases remain visible, without pass/fail labels. |
| 4. Defend conclusion | Prototype did not beat baseline on this measured outcome; explain its failures. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

## State what the evidence can and cannot support

- An exhibition should make the **evidence** inspectable. A polished **demonstration** is not a substitute for the reserved-case record.

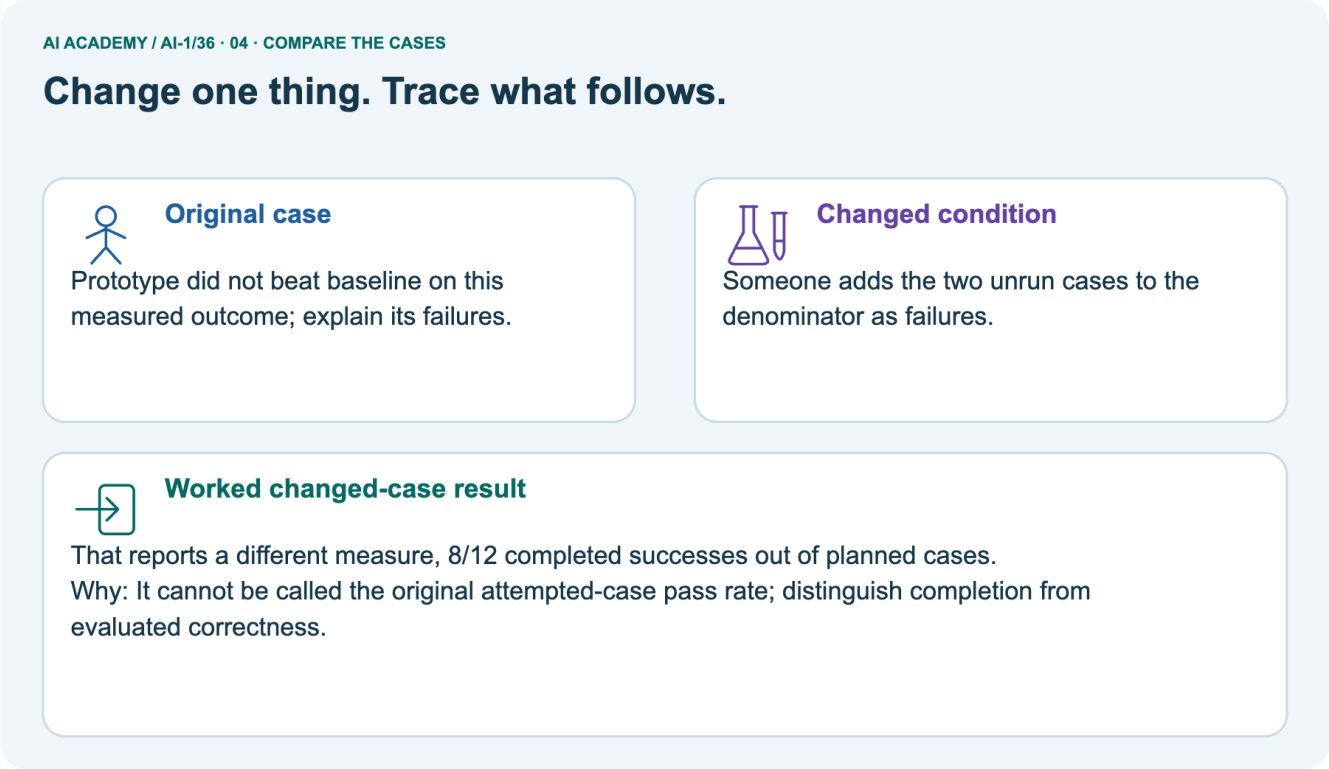
  The shared cases allow a direct descriptive comparison of pass rates; missing timing data prevents a claim that the prototype is faster. Unrun cases must not be treated as successes.

- Read the labelled picture: Write one defensible success statement and one claim the **evidence** cannot support.

## Predict a changed denominator

- Before opening the changed-denominator picture, suppose someone reports eight successes out of all twelve planned cases instead of the ten attempted cases. What different measure does 8/12 describe? Why must the two unrun cases remain distinguishable from observed failures?

### ▼ Check completion versus attempted-case correctness




Follow the labelled picture with the matching explanation. Open the vector diagram

## Keep the classroom-impact limit

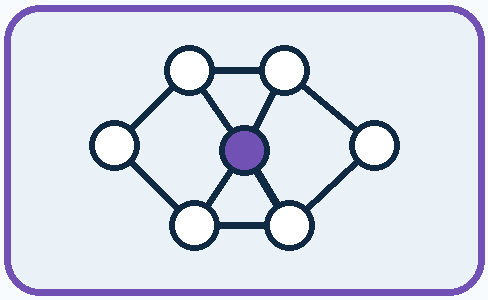
This fictional evaluation does not demonstrate classroom learning impact or performance on every future request.

## ▼ Earlier wording, alternatives and picture transcripts

WORKED EXAMPLE

### Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



CASE A team presents the Classroom Sorter, traces a new item live, shows test **evidence**, explains one failure, and states when a human must decide.

STEP 1 — OBSERVE fresh item and documented system

STEP 2 — TRACE execute and explain each stage under questions

STEP 3 — CONCLUDE result, **evidence**, **limitation**, and responsible next action

MODEL REASONING A complete **defense** traces **input**, **process**, and **output**; distinguishes rules from learned parts; presents normal and edge tests; explains one controlled improvement; states privacy, fairness, uncertainty, and human-review boundaries; and handles a fresh case independently.

DO NOT OVERCLAIM Confidence, decoration, or memorized vocabulary cannot replace a fresh trace and **evidence**-based answers.

What new **evidence** could change this answer?

### A closer explanation

Here is a fictional example from a different project, not your class results. A candidate has 6 full passes from 8 attempted cases; the **baseline** has 8 from the same 8. Two more planned cases were not run.

The candidate passed 75 percent of attempted cases; the **baseline** passed 100 percent. Neither has **evidence** for the two unrun cases. The candidate did not outperform the **baseline** on this measure.

| Evidence | Supported statement | Claim to avoid |
| --- | --- | --- |
| 6 of 8 full passes | Six attempted cases met the full rule | It works for every request |
| Both methods pass the same cases | They tie on this measured result | Automation must be better |
| A time was not measured | Speed is unknown | The candidate is faster |
| Only fictional catalogue data | Results concern these test conditions | People will enjoy every suggestion |

Explain each failure and its consequence instead of hiding it inside a total. Even perfect results on a small reserved set do not establish reliability in every situation.

Our paper adviser applies human-**written rule**s; it has not learned a model from **training** examples. It checks compatibility, not enjoyment, popularity or real material availability.

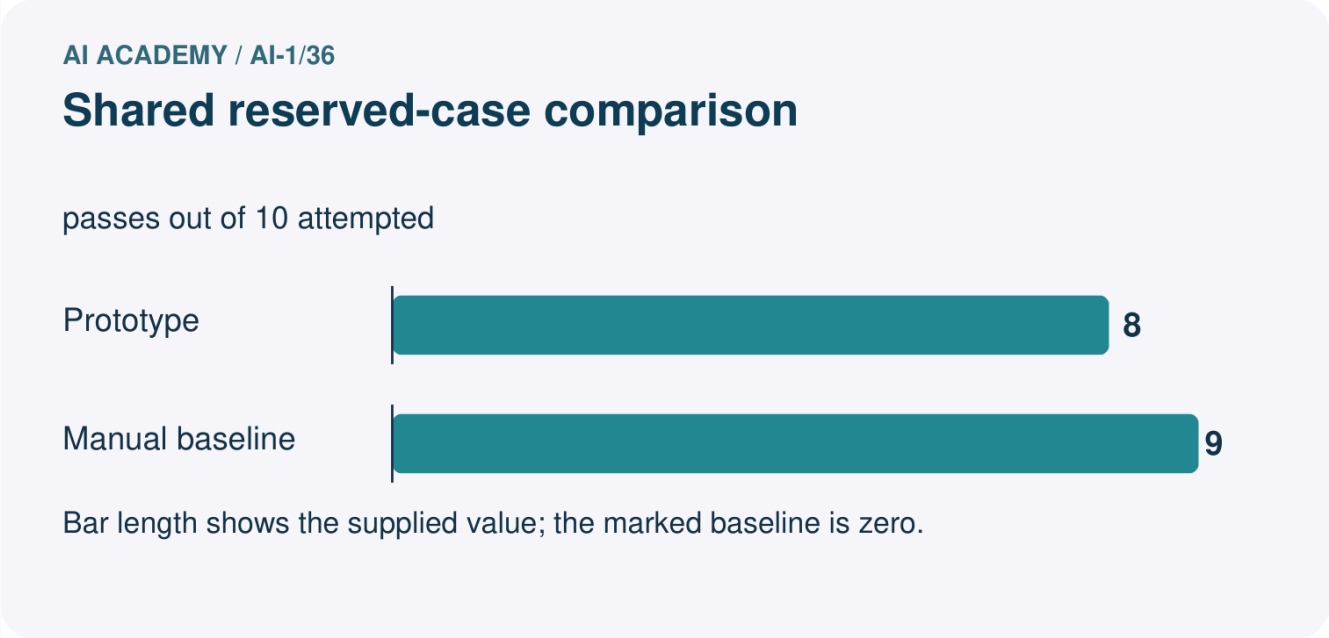
> **In the fictional example, state the candidate and baseline results, explain how the two unrun cases should be reported, and give one useful next action. Why is “the adviser is ready for every classroom” unsupported?**

### Learn from the worked case

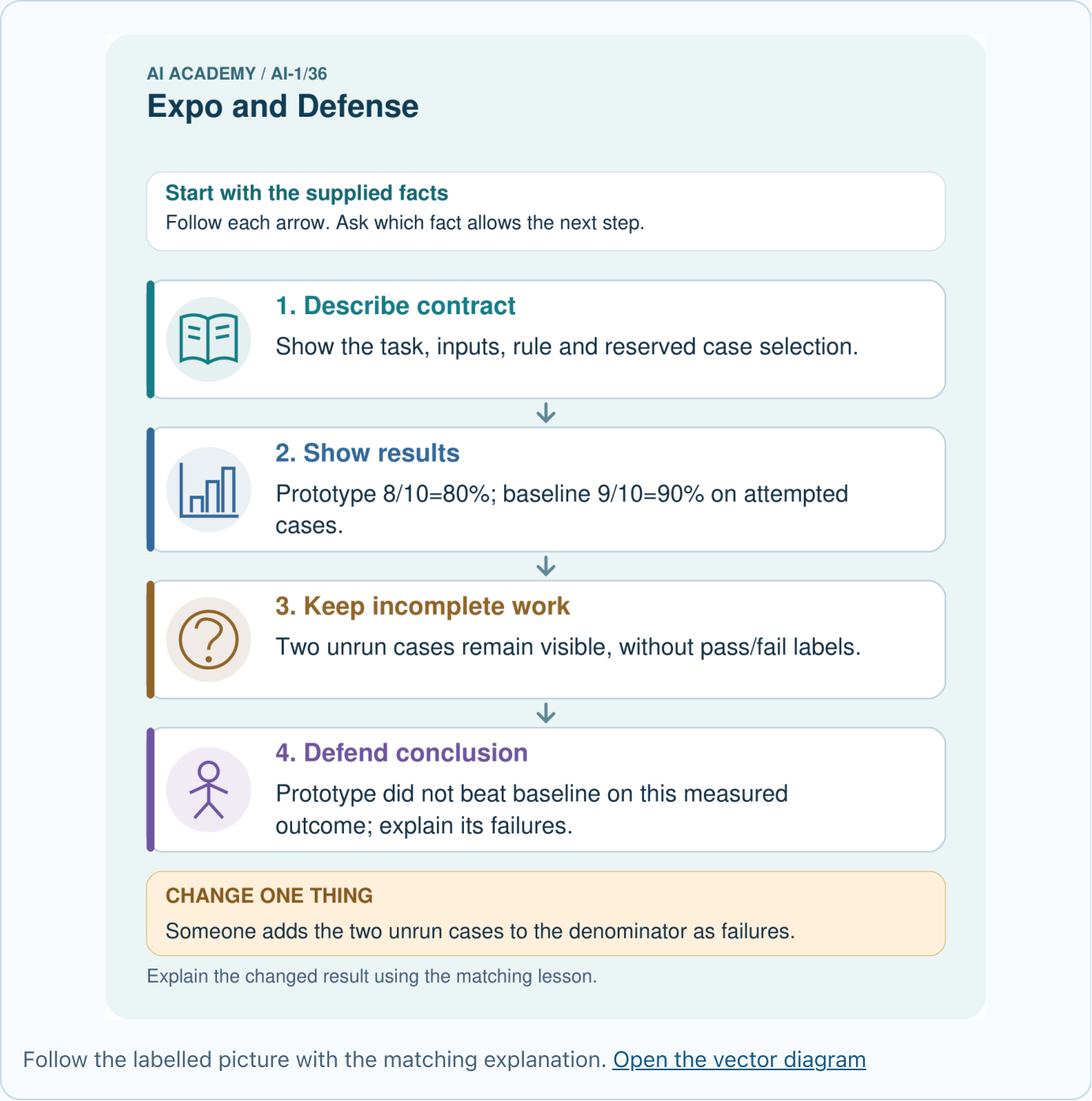
Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

## A pictorial worked case: Expo and Defense

### The facts we will use




Follow the labelled picture with the matching explanation. Open the vector diagram




Follow the labelled picture with the matching explanation. Open the vector diagram

### Why this result follows

### What this example does not establish

Change one thing: Someone adds the two unrun cases to the denominator as failures. Predict the result before checking the explanation. What remains the same?

# Archive the result before any later repair

> A revised version cannot claim its already-seen cases as unseen evaluation.

## Keep evaluated versions and original results

Archive the evaluated candidate and its original results. If you repair it after evaluation, give it a new version label and use revealed cases as development checks. Do not relabel them as unseen.

Ask your teacher for fresh reserved cases only after the revision is fixed. Do not erase the first result.

## State a limited conclusion and justified next action

Write one conclusion limited to the actual conditions tested and one specific next step. It may be better to keep the manual **baseline**. Grade 7 will turn carefully stated procedures into programs and examine data more closely.

Carry forward the habits of validating **input**s, recording **evidence** and keeping people in control.

## Connect actual observations, mechanism and authority

> **Name one observation from your own evaluation, one limit on your conclusion and one next action. Explain why your adviser is a rule-based program design rather than a trained model, and why passing tests does not give it permission to book an activity.**

## Repair both tempting claims

- CLAIM A A successful demo proves every case will work.
- CLAIM B Admitting a **limitation** lowers the quality of the project.

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

▼ Earlier wording, alternatives and picture transcripts

COMPARE

### Same Result, Different Reason

A comparison helps reveal the true concept boundary.

| Case | Evidence | What it shows |
|---|---|---|
| Polished poster | communication aid | not proof alone |
| Fresh trace | unrehearsed application | transfer evidence |
| Failure log | honest testing record | engineering evidence |
| Human gate | responsible boundary | use condition |

### Repair two tempting claims

Repair:

Repair:

### A closer explanation

---

MAIN LESSON · SECTION 8

## Practise an evidence defence with human control

> Prepare a short explanation supported by your actual records.

## Explain the saved rules, trace and any justified repair

Prepare a short explanation using your saved instructions, one complete trace, the final comparison, and a **limitation**. State the problem and whose choice the adviser supports.

Explain validation, both matching conditions, first-fit order, reset and stopping. Show the exact instruction behind your trace. Describe any repair with its old **evidence**, or say honestly that no repair was needed.

## Keep inputs minimal and suggestions under human choice

Use only fictional requests and the two necessary **input** fields. Do not collect names, diagnoses or reasons a pupil prefers quiet. A quiet request is enough.

A person may decline the suggestion; a teacher checks real materials and local rules before an activity. A successful **output** grants no permission to book or publish.

## Use the two-minute presentation order

1. For a two-minute partner presentation, use this order: problem and rule; trace; candidate versus **baseline** results; failure or **limitation**; next action. Partners swap speaker and listener roles.

   Each person must explain one trace independently. The listener asks which recorded **observation** supports the main claim; attractive slides cannot substitute for that **observation**.

## Keep the final evaluation record in W2

Use workbook W2 for the connected example “Evaluating and explaining your project”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

| Plan | Do | Check |
| --- | --- | --- |
| What is given? Which rule or procedure applies? | Record each intermediate step. | Does the result follow? What remains unknown? |

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

## Challenge the polish-proves-performance claim

- Use the worked case for Evaluating and explaining your project. Proposed explanation: “A polished presentation proves the system works”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

## Return to the exhibition-board conclusion

Write one defensible success statement and one claim the **evidence** cannot support.

This is supported practice. Keep the separately named independent case for an attempt before feedback.

### ▼ Earlier wording, alternatives and picture transcripts

GUIDED PRACTICE

**Try It, Check It, Explain It**

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Understanding is shown by explanation, **transfer**, testing, and limits.

Use a fresh example to explain **demonstration**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: Admitting a **limitation** lowers the quality of the project.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

**A closer explanation**

**Practise with less help: Keep the final evaluation record**

**Check your reasoning**

**Explain the picture**

> **My explanation using a pictured step and the given facts**

---

MAIN LESSON · SECTION 9

# Run the final evaluation and evidence defence

> Keep the teacher's reserved inputs and key separate until the appropriate stage.

## Archive, receive inputs and hold the candidate fixed

1. Final paper evaluation and defence: first archive the complete candidate instructions, label, development records and known limitations. Then receive the teacher's fresh labelled activity catalogue and requests without the key. Apply the existing whole-minutes 0–30, Yes/No quiet, inclusive time, both-condition and first-fit rules. Use the same assigned data for candidate and manual baseline. Keep the candidate unchanged during the evaluation.

## Record actual comparisons and defend a bounded claim

1. For each assigned case save both methods' actual outputs, reasons and traces before checking the key. Count full passes out of attempted cases and list unrun cases separately. Present the problem, one actual trace, both methods' results, a retained failure or limitation and a justified next action; each learner explains a trace independently. If you later repair the candidate, preserve this evaluation and label revealed-case reruns as development evidence for the revision. Do not invent results or claim timing you did not measure.

2. Keep a normal case and an edge case clearly separated.

3. Record what happened—even when it does not match your **prediction**.

4. Explain the result, one **limitation**, and the next useful test.

## Retain failures, unrun cases and next action

| Case | Prediction | Observed result | What it means |
|---|---|---|---|
| Normal | | | |
| Edge | | | |

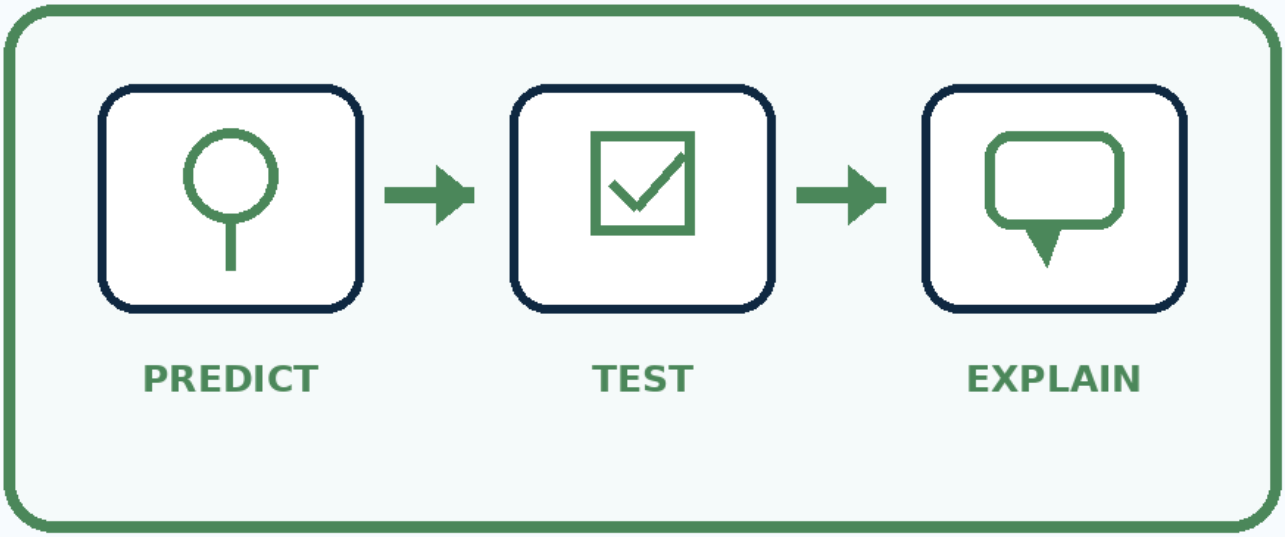
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep "not run" if you only traced the procedure.

### ▼ Earlier wording, alternatives and picture transcripts

EXPLORER LAB

#### Responsible Classifier Expo

Predict first, test honestly, and explain what the **evidence** means.




State your **prediction** before the result is known.

Complete the task: Demo the system, solve a fresh case, and answer why it may fail.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

**Plan → try → check**

---

MAIN LESSON · SECTION 14

## Try alone: judge the different library adviser

> Audit the supplied counts, unsupported claims and meaning of a rerun.

### The independent six-case comparison

A fictional library adviser and its manual **baseline** face the same 6 reserved cases. The candidate has 4 full passes; the **baseline** has 5. One additional planned case is unrun by both. No speed data was collected.

The team says “ours is faster and more reliable everywhere” and repairs it after seeing the failures. Explain what the counts support, two unsupported claims and how to label a rerun of the revealed cases.

## Keep your first attempt

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh **transfer**.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

## Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

### ▼ Earlier wording, alternatives and picture transcripts

#### Independent case: Judge a different project independently

> **My independent explanation, trace and limits**
>
> _____
>
> _____
>
> _____
>
> _____

#### Week 36 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

At the expo, a team says our sorter never fails after testing four cards. Replace the claim with an **evidence**-based statement and name a next test.

My answer, with a trace, calculation or supporting **evidence**

**Write or draw your reasoning and evidence.**

### Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

**Write or draw your reasoning and evidence.**

### Connect to your project

Present the final Classroom Sorter portfolio and complete the AI-1 certification **defense**.

The **evidence** I will keep

**Write or draw your reasoning and evidence.**

### Transfer practice from the original programme

Defend a sorter with 10/12 routine successes and 2/4 mixed-material successes. Report overall and separate results, show one failure and propose a justified next step. What claim would overstate the **evidence**?

**Explain the mechanism, evidence and one limit.**

**Use feedback to improve**

---

MAIN LESSON · SECTION 11

# Present the earlier sorter portfolio with its own evidence

> Keep activity evaluation results distinct from sorter records.

## Connect the two project contexts

The connected final sequence evaluates the paper activity adviser. This section assembles the evidence from your earlier Helpful Classroom Sorter portfolio; it does not merge the two projects or require a second final build. Label which project produced each trace and result. An honest limitation or decision to retain the manual method is valid evidence of understanding.

### This week’s sorter portfolio

THIS WEEK'S CHECKPOINT Present the final Classroom Sorter portfolio and complete the AI-1 certification **defense**.

Present the Helpful Classroom Sorter’s versions, test records, **baseline**, failures and next investigation as an **evidence** portfolio.

## Defend the actual decisions and limits

- What real classroom need does this serve?

- What information is necessary—and what should stay private?
- What happens inside the **process**?
- What normal and edge cases will you test?
- When should it say not sure or ask a person?

## Use evidence rather than decoration

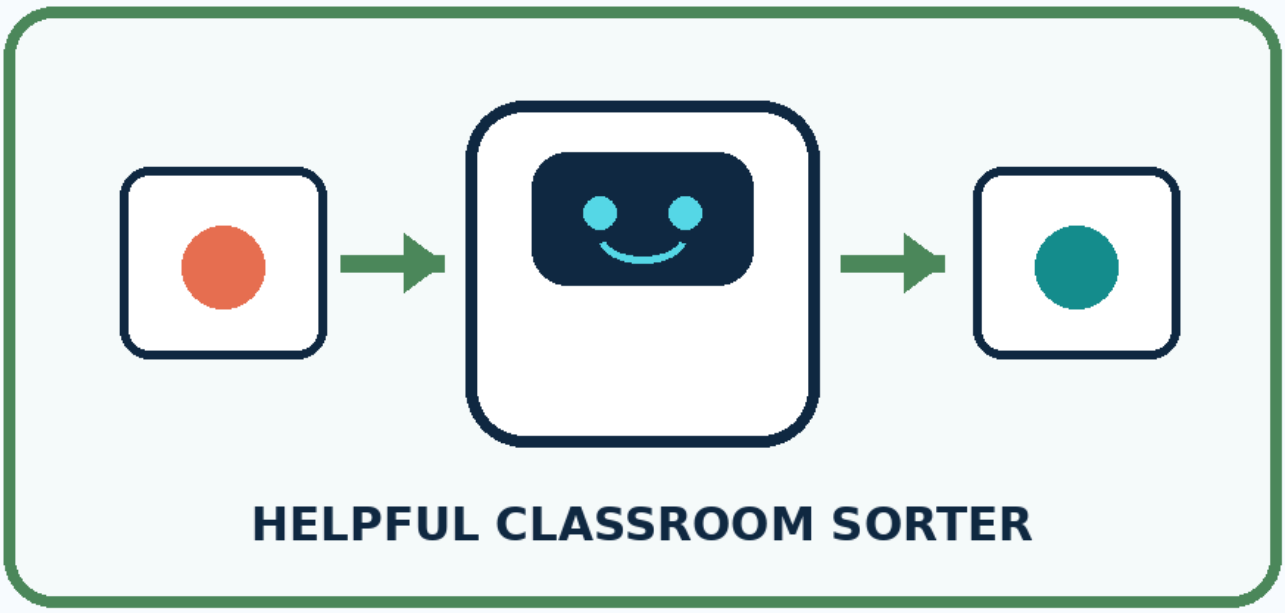
RESPONSIBLE DESIGN Confidence, decoration, or memorized vocabulary cannot replace a fresh trace and **evidence**-based answers.

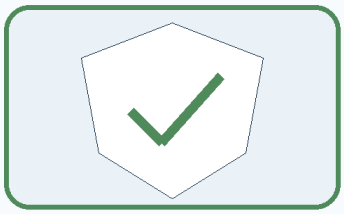
▼ **Earlier wording, alternatives and picture transcripts**

PROJECT

### Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.





My project decision:

**Evidence for your project**

---

MAIN LESSON · SECTION 16

# Finish: defend one observation, limit and next step

> Close your notes and explain what your own evidence supports.

## Your exit explanation

- Close your notes. Explain this goal in your own words: Evaluate a saved candidate and a manual **baseline** fairly

**My deciding step, evidence and one remaining question**

## Plan a changed-case return and next-level support

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

# Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

## Your quick word guide

> Use the words to explain a particular step, record or limit.

### Core vocabulary

| Word | Meaning | My example |
|---|---|---|
| demonstration | live showing of system behavior | |
| defense | evidence-based explanation under questions | |
| transfer | applying understanding to a fresh case | |
| limitation | condition where performance or use is restricted | |

### Terms for the connected explanation

| Word | Meaning |
|---|---|
| Final evaluation | Testing a saved candidate under stated conditions after development |
| Baseline | A comparison method tested under the same conditions |
| Evidence | Recorded observations that support or challenge a claim |

| Word | Meaning |
| --- | --- |
| Defence | An explanation of a decision using evidence and reasoning |

## Repair a claim using a word

- CLAIM TO REPAIR A successful demo proves every case will work.
- Use one vocabulary word and one piece of **evidence** to repair it:

### ▼ Earlier wording, alternatives and picture transcripts

VOCABULARY

#### Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.












### Misconception detective

WORD POWER A strong definition helps you decide something in a new case.

### Words used in the closer explanation

▼ Choose a stronger thinking challenge

REFERENCE · SECTION 10

# Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

## Pick your next step

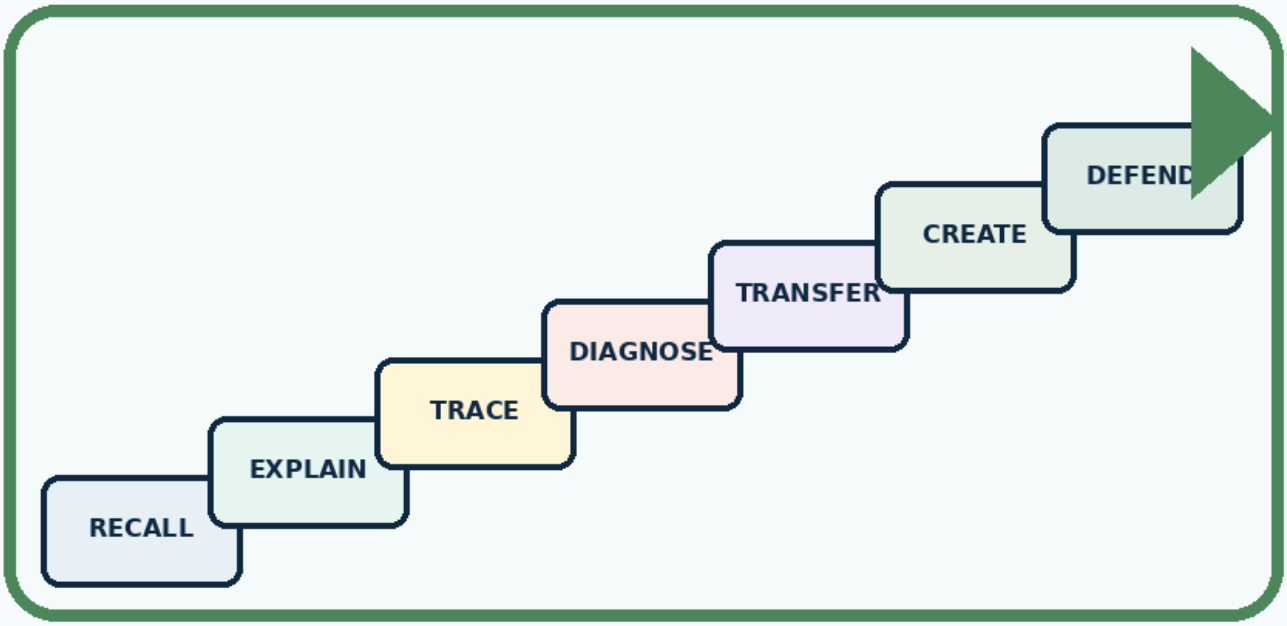
- RECALL Define **demonstration**.
- EXPLAIN Understanding is shown by explanation, **transfer**, testing, and limits.
- TRACE Show the **input**, **process**, and **output**.
- DIAGNOSE Repair: A successful demo proves every case will work.
- **TRANSFER** Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

▼ Earlier wording, alternatives and picture transcripts

PRACTICE LADDER

### Climb from Recall to Defense

Each step asks for a stronger kind of understanding.




RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

▼ **Reference: check your explanation**

REFERENCE · SECTION 12

# Reference: check your explanation

> Use after practice. Judge a different project independently is the selected independent case.

## Check the evidence and its limit

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.

- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

## Record support

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

### ▼ Earlier wording, alternatives and picture transcripts

MASTERY

#### Checklist for explaining this week's idea

Use after practice. Judge a different project independently is the selected independent case.




EXPLANATION CHECK Compare the same attempted cases, preserve unrun plans and original evaluations, and support every claim with actual measured **evidence**.

Independent **evidence**:

SELF-CHECK A complete **defense** traces **input**, **process**, and **output**; distinguishes rules from learned parts; presents normal and edge tests; explains one controlled improvement; states privacy, fairness, uncertainty, and human-review boundaries; and handles a fresh case independently.

NEXT BRIDGE Carry today's **evidence** habit into: the next Academy level.

▼ **Extra practice: Defend a claim with a complete trace**

REFERENCE · SECTION 13

## Extra practice: Defend a claim with a complete trace

> Try the question before opening its worked explanation.

### Read the supplied case and question

The expo poster says always correct after six classroom passes.

Write a defensible claim.

### Your explanation

- Which detail supports the answer, and why does it matter?

**Write or draw your reasoning and evidence.**

## ▼ Check after trying

Report six specified passes with conditions and limits. Show an **input**, decision trace, **output** and test record, including an uncertain case.

| Evidence or stage | Value or result |
| --- | --- |
| Input | Show a specified test object |
| Decision | Explain its path |
| Output | Show the tray or review result |
| Evidence | Report six classroom passes and limits |

## ▼ Earlier wording, alternatives and picture transcripts

### Week 36 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

### The situation

### The question

### Worked explanation

### Explain it in your own words

### Trace the supplied case visually

### Defend a claim with a complete trace

The expo poster says always correct after six classroom passes.

Write a defensible claim.

Report six specified passes with conditions and limits. Show an **input**, decision trace, **output** and test record, including an uncertain case.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

# Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

## Targets and evidence

| By the end, I can | How I will show it |
|---|---|
| Evaluate a saved candidate and a manual baseline fairly | Explain the deciding step, show a trace or test, and state one limit. |
| Defend a conclusion using results, failures and limits | Explain the deciding step, show a trace or test, and state one limit. |
| Explain the rules, privacy choices and human control in your project | Explain the deciding step, show a trace or test, and state one limit. |

## Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional paper sorter team prepares an exhibition **evidence** board.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

# Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

## First supported question

- A fictional paper sorter team prepares an exhibition **evidence** board.
- Which labelled step matches this detail? Prototype did not beat **baseline** on this measured outcome; explain its failures.
- 1. Keep incomplete work
- 2. Describe contract
- 3. Show results
- 4. Defend conclusion
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Practice 1: explain the deciding evidence

**My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Defend conclusion Prototype did not beat **baseline** on this measured outcome; explain its failures. An exhibition should make the **evidence** inspectable.

A polished **demonstration** is not a substitute for the reserved-case record. The shared cases allow a direct descriptive comparison of pass rates; missing timing data prevents a claim that the prototype is faster.

Unrun cases must not be treated as successes.

## Changed-case question

- Changed-case practice: Someone adds the two unrun cases to the denominator as failures. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. That reports a different measure, 8/12 completed successes out of planned cases.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed case**

After your attempt, compare with the supported explanation: That reports a different measure, 8/12 completed successes out of planned cases.

It cannot be called the original attempted-case pass rate; distinguish completion from evaluated correctness.

## Repair and save your work

Review Improve One Cause (ai-1/35). Goal: Repair a located defect without changing the agreed requirements. Short practice: Explain why “our repaired version now picks something” is not enough to show success at 6..

Return to this week and explain what you changed.

Review Test Normal Cases (ai-1/33). Goal: Plan routine cases with expected results derived from the requirements. Short practice: Why is none correct for 8 but incorrect for 6? Use both capacity cards..

Return to this week and explain what you changed.

Review Choose **Input**s and **Output**s (ai-1/30). Goal: Turn the project outcome into precise **input** and **output** requirements. Short practice: Trace requests 10 and blank. Why should they not receive the same explanation?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

### ▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

#### Repair a missing building block

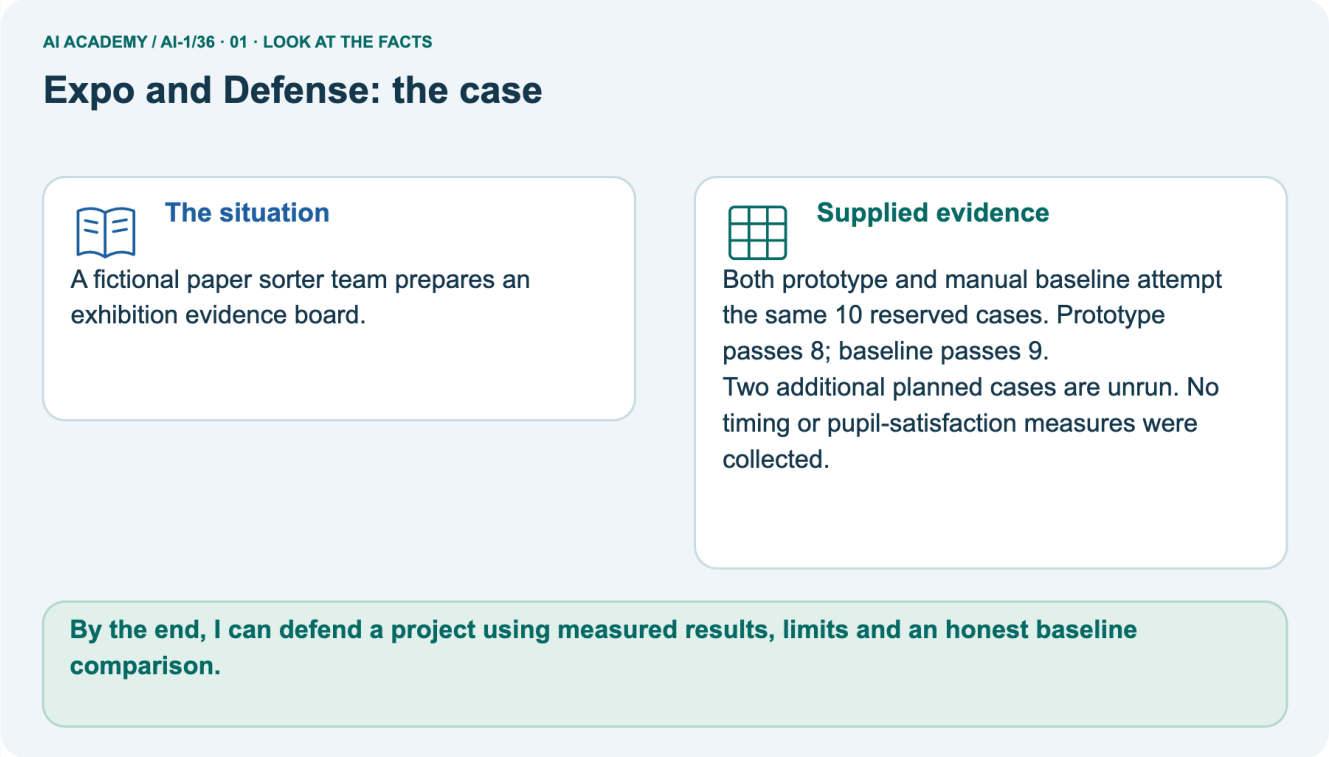
#### Keep your completed work

### ▼ Reference: the six-picture journey

# Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

## 01. Expo and Defense: the case



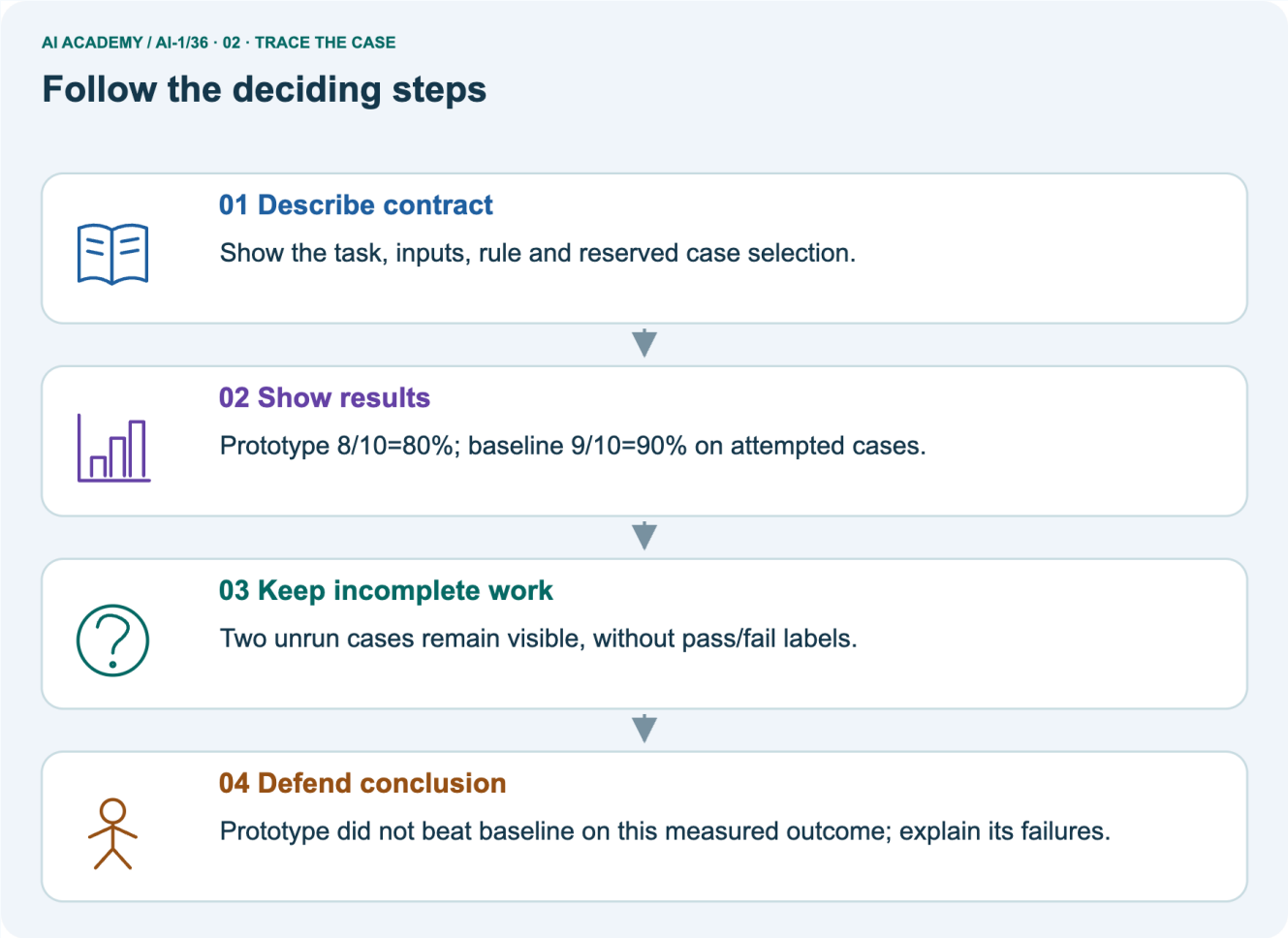

Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional paper sorter team prepares an exhibition **evidence** board. Supplied facts: Both prototype and manual **baseline** attempt the same 10 reserved cases. Prototype passes 8; **baseline** passes 9.

Two additional planned cases are unrun. No timing or pupil-satisfaction measures were collected.

Point to the given fact that matters most. Explain why it matters.

## 02. Follow the deciding steps

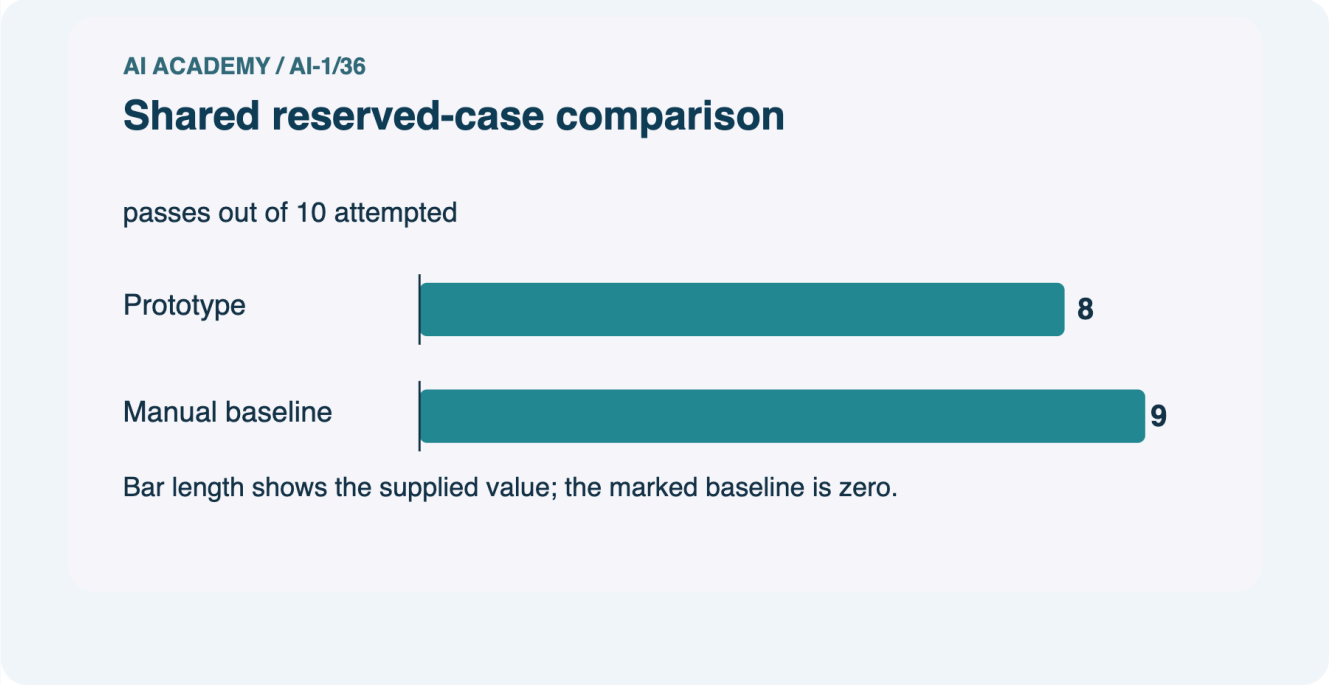



Follow the labelled picture with the matching explanation. Open the vector diagram

Describe contract: Show the task, **input**s, rule and reserved case selection. → Show results: Prototype 8/10=**80%**; **baseline** 9/10=90% on attempted cases. → Keep incomplete work: Two unrun cases remain visible, without pass/fail labels. → Defend conclusion: Prototype did not beat **baseline** on this measured outcome; explain its failures.

Point to each numbered stage and explain which recorded **input** it uses.

## 03. Shared reserved-case comparison

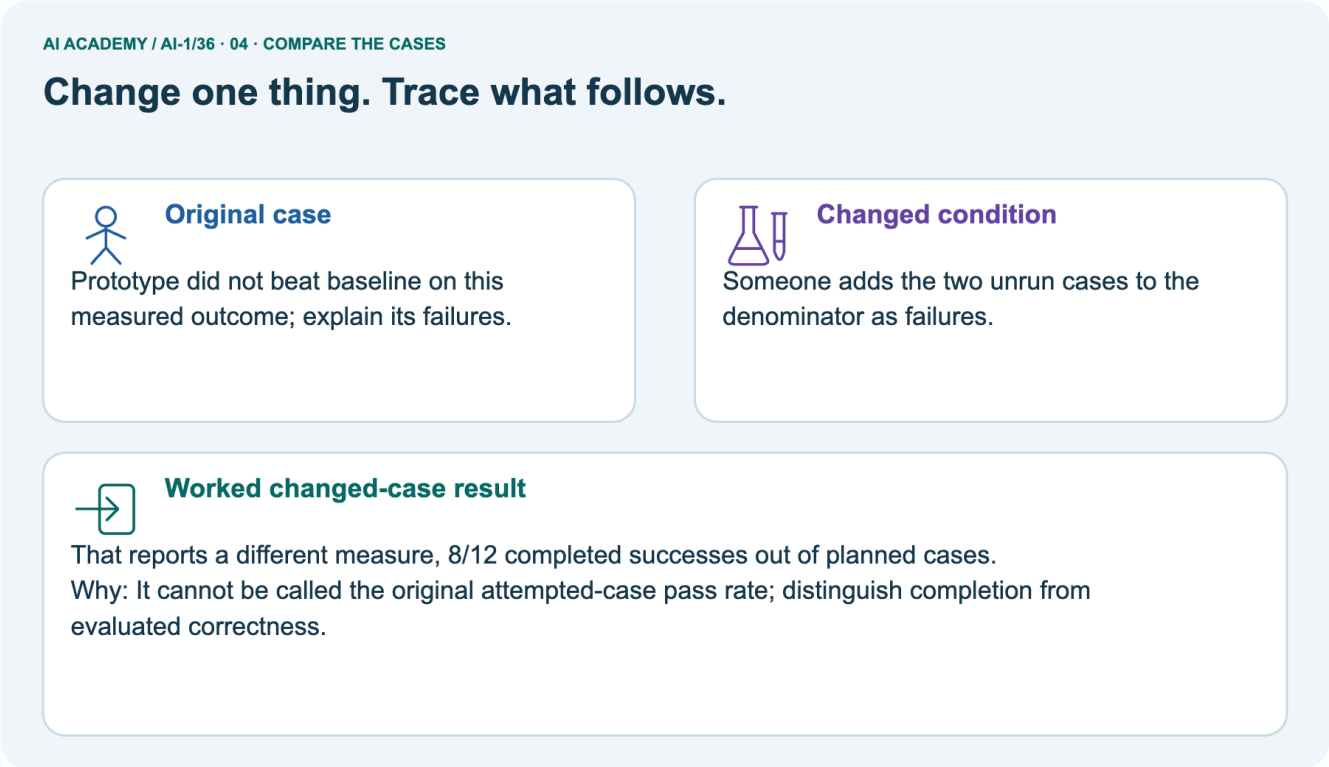



Follow the labelled picture with the matching explanation. Open the vector diagram

Shared reserved-case comparison. Units: passes out of 10 attempted. Prototype: 8. Manual **baseline**: 9.

Read the labels and values. Explain the relationship the model shows.

## 04. Change one thing. Trace what follows.



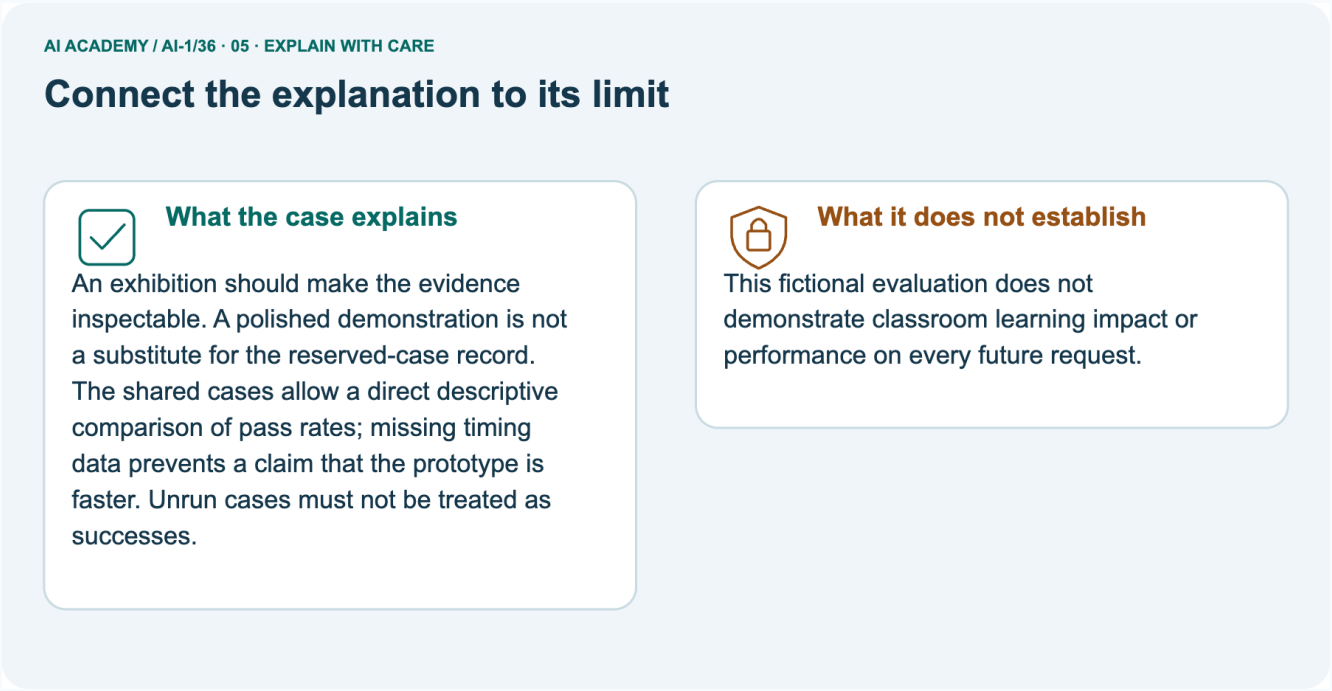

Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Prototype did not beat **baseline** on this measured outcome; explain its failures.
Changed condition: Someone adds the two unrun cases to the denominator as failures.

Expected result: That reports a different measure, 8/12 completed successes out of planned cases. Reason: It cannot be called the original attempted-case pass rate; distinguish completion from evaluated correctness.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

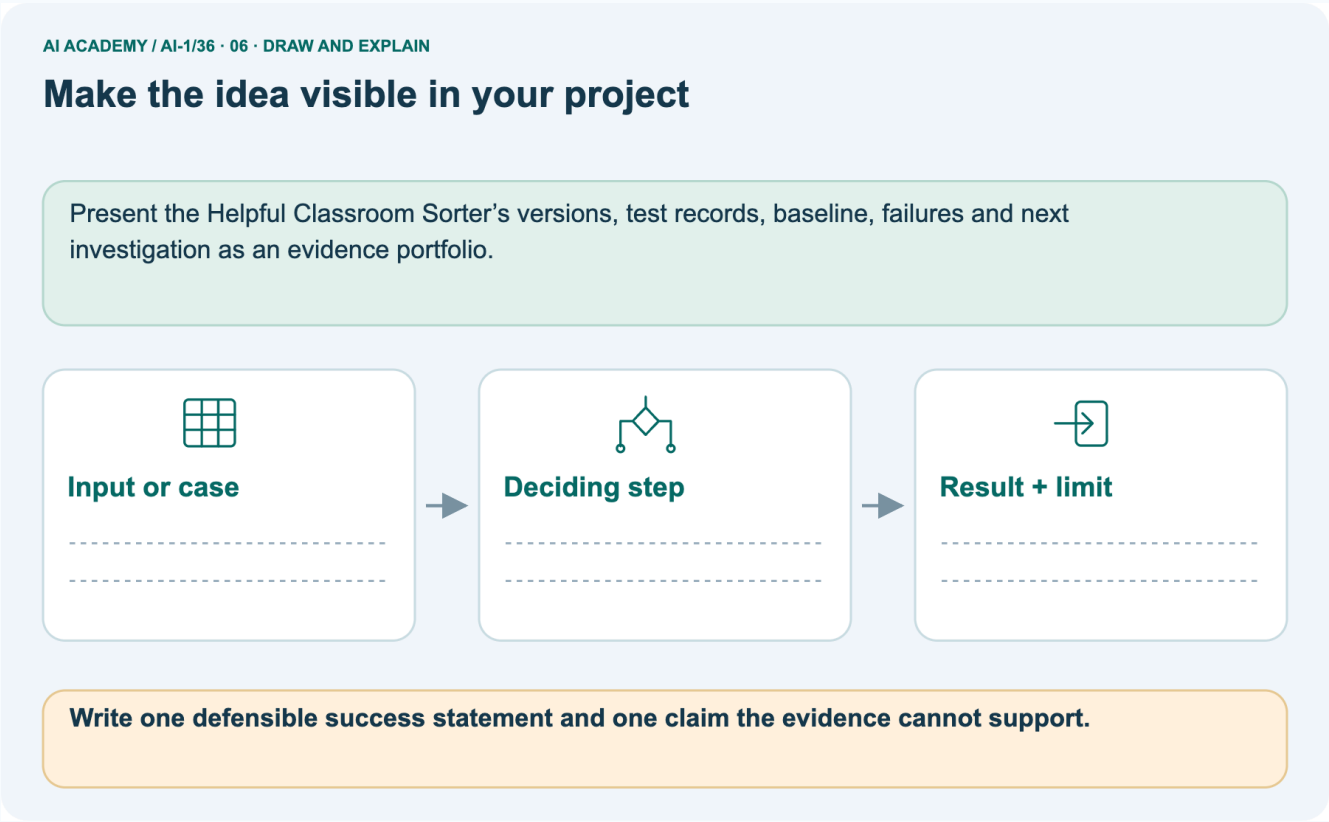
An exhibition should make the **evidence** inspectable. A polished **demonstration** is not a substitute for the reserved-case record.

The shared cases allow a direct descriptive comparison of pass rates; missing timing data prevents a claim that the prototype is faster. Unrun cases must not be treated as successes.

Limit: This fictional evaluation does not demonstrate classroom learning impact or performance on every future request.

Explain the case, then point to a limit on the claim.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

Present the Helpful Classroom Sorter’s versions, test records, **baseline**, failures and next investigation as an **evidence** portfolio. Draw your **input**, deciding step, result and limit.

Supported question: Write one defensible success statement and one claim the **evidence** cannot support.

Draw your project trace and justify the deciding step using **evidence**.

## ▼ Earlier wording, alternatives and picture transcripts

**01. Expo and Defense: the case**

**02. Follow the deciding steps**

**03. Shared reserved-case comparison**

**04. Change one thing. Trace what follows.**

**05. Connect the explanation to its limit**

**06. Make the idea visible in your project**

## ▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.